

Do Hedge Funds Supply or Demand Liquidity?

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Abstract

Regressing hedge funds' returns on returns to a long-short contrarian trading strategy, a measure of the returns from providing liquidity, we find that hedge funds typically supply liquidity in the stock market. In the cross-section, strict redemption restrictions and large fund size increase funds' propensity to supply liquidity. In time-series, poor market liquidity and good funding conditions increase funds' propensity to supply liquidity. Although the hedge funds typically supply liquidity, during crises they demand liquidity. We also find that increases in the amount of speculative capital improve market liquidity and reduce the amount of short-term return reversals and volatility.

Key words: Hedge Funds, Speculative Capital, Liquidity, Returns from Providing Liquidity, Volatility

JEL Classifications: G12, G23

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1. Introduction

It is well established that there are significant stock return reversals at one-week and one-month horizons, see e.g., Jegadeesh (1990) and Lehmann (1990). The finance literature, e.g., Jegadeesh and Titman (1995), links the short-term return reversals with imperfect liquidity in financial markets. In illiquid markets, return reversals emerge from transitory investors' portfolio imbalances due to imperfect risk bearing ability of market makers (Grossman and Miller, 1988). For instance, when an investor with an urgent need to sell any given stock arrives at the equity market, the stock price must temporarily decline below its fundamental value to induce market makers to provide liquidity, i.e., to become counterparties to the investor's trade and clear the market. Later, when new investors arrive at the market, prices revert to fundamentals. In illiquid markets, the market makers thus act as contrarian traders and earn returns from supplying liquidity, or immediacy, to other investors (Hendershott and Seasholes, 2007).

In this paper, we study hedge funds' role in the supply and demand of liquidity. Hedge funds can act as contrarian traders and, along with the market makers, participate in the supply of liquidity in the stock market. Alternatively, it can be that hedge funds typically demand liquidity in the stock market as many of their equity market strategies, such as momentum, require frequent trading.¹

Our proxy for the returns to hedge funds from providing liquidity in the stock market is the return to a contrarian zero-investment long-short trading strategy that buys stocks with positive expected weekly excess return, calculated using past estimates of short-term return reversal and past returns, and sells short stocks with negative expected excess return. Our approach to estimating the returns from providing liquidity is closely related to that in Khandani and Lo (2011a), Nagel (2012), and Rinne and Suominen (2011). In order to investigate whether hedge funds typically supply or demand liquidity, we simply regress hedge fund returns on the returns from providing

¹ See Scholes (2004) for discussion on hedge funds' role in the supply of liquidity.

liquidity and controls. If the regression coefficient for the returns from providing liquidity is significantly positive, we conclude that the fund typically supplies liquidity. If the regression coefficient is significantly negative, it indicates to us that the fund is a liquidity demander.²

Our main finding is that hedge funds typically supply liquidity in the stock market. The average exposure of hedge funds to our returns from providing liquidity series is positive and nearly 6% of all hedge funds individually have a statistically significant positive exposure to the returns from providing liquidity series. Strikingly, if we consider the possibility that it may take time for hedge funds to trade, and look at the hedge funds' exposure to our returns from providing liquidity portfolio during a ten day forward window, we find that as many as 24% of all hedge funds are statistically significantly involved in the supply of liquidity in the stock market. In contrast, only 6% of the funds are found to significantly demand liquidity during these time periods. Given this, our evidence that hedge funds typically supply liquidity is very strong.

We find that in the cross-section the funds that offer infrequent redemption possibilities and large funds have a higher propensity to supply liquidity compared to other funds. There are also time-varying factors that affect the hedge funds' propensity to supply liquidity: According to our results, hedge funds reduce their supply of liquidity when market liquidity is high, and presumably

² Following Nagel (2012) and Rinne and Suominen (2011) we refer to trading profits from contrarian trading strategies as returns from providing liquidity. Conceptually our returns from providing liquidity correspond with the returns from providing immediacy in Grossman and Miller (1988). The returns from providing liquidity should be distinguished from the illiquidity premium and the liquidity risk premium, which have been defined as the excess return from holding illiquid stocks (Amihud and Mendelson, 1986) and the excess returns from stocks that are exposed to aggregate shocks in the level of liquidity (Pastor and Stambaugh, 2003, Acharya and Pedersen, 2005, and Sadka, 2006), respectively. In time-series, the returns from providing liquidity are likely to vary with the level of market liquidity as the degree of return reversal and liquidity are negatively related, see e.g., Avramov et al. (2006). In fact, return reversal is commonly used to measure market illiquidity, see e.g., Pastor and Stambaugh (2003).

the returns from providing liquidity are low, and, consistent with Nagel (2012), when credit market is tightening and the funds' ability to supply liquidity is constrained.

Although hedge funds typically supply liquidity in the stock market, we find evidence from rolling regressions that during liquidity related crises, notably the 2007 quant crisis, the 2008 financial crisis, and the 2011 collapse of derivatives broker MF Global, hedge funds demand liquidity. The finding that hedge funds are more prone to demand liquidity during the crises is consistent with the findings in Nagel (2012), Ben-David et al. (2012), and Cella et al. (2013). Nagel (2012) documents that the increase in the VIX index during the financial crisis was associated with an increase in the returns from providing liquidity, suggesting either an abnormally large demand for liquidity in the market, or market participants' impaired ability to supply liquidity during the crisis. Ben-David et al. (2012), in turn, document that hedge funds, while Cella et al. (2013) that short-term traders in general, were actively selling during the market fall associated with the financial crisis. The evidence in Cella et al. (2013) suggests that this selling was liquidity demanding also within the cross section of stocks, as the stocks held by short-term investors prior to the crisis had more severe price drops during the crisis, followed by larger reversals compared to other stocks. One contribution of our paper is to provide evidence that the hedge funds' tendency to demand liquidity during crises is not just specific to the 2008 financial crisis. In addition, although this is largely explained by hedge funds' demand for liquidity during the 2008 financial crisis, we find evidence that the most extreme cross-sectional return reversals are associated with stocks where hedge funds, at that point in time, demand liquidity.³

³ Note that in our paper we only look at the supply and demand of liquidity within the cross-section of stocks. We ignore the more macro level demand and supply of liquidity that can occur between asset classes, for instance if hedge funds are net sellers of shares and net buyers of bonds. The demand and supply of liquidity between asset classes is beyond the scope of the current paper. For evidence on hedge funds' exit from the equity market at an aggregate level during the liquidity crises, see Ben-David et al. (2011).

Given the evidence that hedge funds typically supply liquidity in the stock market, we study how the capital flows into the hedge fund industry affect the stock market. Consistent with Brunnermeier and Pedersen (2009) and Gromb and Vayanos (2010), we find that increases in the amount of speculative capital that hedge funds employ, either in assets under management or debt, improve market liquidity and reduce the amount of short-term return reversal and volatility. Finally, we provide evidence that the introduction of automated quote dissemination (autoquote) at the NYSE, fostering high frequency algorithmic trading, has changed hedge funds' behavior. Under autoquote, the hedge funds both enter and exit later their liquidity providing trades.⁴

Previous empirical research on hedge funds' role in the supply of liquidity is scarce. The only other paper that focuses on the hedge fund's role in supplying liquidity that precedes our work is, to our knowledge, Aragon and Strahan (2012). They show that following Lehman Brothers' bankruptcy, stocks traded by the Lehman-connected hedge funds experienced greater declines in market liquidity than other stocks. This is evidence that following their prime broker's bankruptcy the hedge funds connected to Lehman Brothers were less able to supply liquidity than at other times, affecting market liquidity. With our methodology we can provide important information on the degree of and time variation in the hedge funds' supply and demand of liquidity, which helps put in context for instance the earlier findings in the literature on the hedge funds' trading activities during the crises, as documented in Aragon and Strahan (2012) and Ben-David et al. (2012). Furthermore, our results provide evidence of the important role of funding liquidity to hedge funds and its relation with market liquidity, in support of the arguments presented in Brunnermeier and Pedersen (2008) and Nagel (2012). Our results on the gradual arrival of the hedge funds' liquidity provision, and how their speed of liquidity provision was altered after the technological changes that increased the popularity of high frequency trading are of interest on their own.

⁴ See Hendershott et al. (2011) for related discussion of how algorithmic trading has improved market liquidity.

Prior to our work researchers have studied mainly the linkages between liquidity, liquidity risk, and hedge fund returns. For instance, Sadka (2010) shows that hedge funds are exposed to unexpected market wide liquidity shocks. He argues that this exposure is due to hedge funds' portfolio allocations, where long positions are commonly taken in more illiquid securities than short positions to benefit from the illiquidity premium. As an example, he refers to the case of convertible arbitrage, where long positions are taken in illiquid convertible bonds, while short positions are taken in more liquid stocks. Second, however, he points out that hedge funds' exposure to market wide liquidity changes could result from the use of trading strategies that require liquidity due to their frequent trading interval, such as the momentum strategy. In all our tests of whether hedge funds supply or demand liquidity, we include the unexpected shocks to liquidity, as measured in Sadka (2010), as a control. It turns out that the two phenomena co-exist. Several hedge funds, or hedge fund style categories as aggregate, have a statistically significant exposure to the returns from providing liquidity. In addition several hedge funds have a significant exposure to unexpected changes in liquidity.

In addition to Sadka (2010), there are a few other papers that present evidence of hedge funds' exposure to the level of liquidity or liquidity risk. For instance, Cao et al. (2012) study whether hedge funds can time liquidity. Getmansky et al. (2004) in turn explore reasons for high autocorrelation in hedge fund returns and argue that exposure to illiquid securities is the most likely explanation. Also Boyson et al. (2010) show evidence that liquidity shocks are the main drivers of hedge fund returns and that liquidity affects the probability of contagion in hedge funds. Gibson and Wang (2013), in turn, show that hedge funds' return predictability (documented by e.g., Avramov et al., 2011) is, at least partially, driven by the funds' exposure to liquidity risk. Other papers in this stream of literature include Aragon (2007), Aragon et al. (2012), Khandani and Lo (2011b), and Teo (2011). Kang et al. (2012) is also related. They study the effect of hedge fund activity on

idiosyncratic volatility. One of their main findings is that the downward trend in the volatility of low idiosyncratic volatility stocks can be attributed to long-short equity hedge fund activity.

The rest of the paper is organized as follows. In Section 2, we describe our method for estimating the returns from providing liquidity. In Section 3, we present our main empirical results regarding hedge funds' propensity to supply liquidity. In Section 4, we analyze the effect of hedge funds' trading on short-term return reversals, liquidity, and volatility. In Section 5, we study the robustness of our results, in Section 6 the case of extreme return reversals, and in Section 7 the effect of introduction of autoquote at the NYSE on hedge funds' supply of liquidity. Section 8 concludes the paper.

2. Measuring the Returns from Providing Liquidity

2.1 DATA

When estimating return reversal patterns, our sample includes all stocks listed in the daily Center for Research in Security Prices (CRSP) file from January 1st, 1990 to December 31st, 2011, which fulfill the following requirements: 1) the security is an ordinary common stock, 2) the company is incorporated in the U.S., 3) the stock is listed in the NYSE or the Amex, and 4) the company's Standard Industrial Classification code is available and it is included in the Fama-French 48 industries, excluding the industry Other. Further, when estimating the returns from providing liquidity, we make additional data restrictions to reduce noise in our estimates. First, we remove from our sample all stocks that belong to the smallest decile of all U.S. incorporated common stocks listed on the NYSE or the Amex. Second, we eliminate penny stocks by removing from our

sample all stocks that have a share price below five dollars. Finally, we require that a stock must have a positive trading volume during each day when a position in the stock is presumably opened.⁵

2.2 SHORT-TERM RETURN REVERSAL AND EXPECTED RETURNS

To estimate the return reversal patterns in excess returns, we perform for each day a cross-sectional regression, in which we regress the stocks' (indexed by i) next 5 days' (one week) excess returns following the close on day t , $R_{i,t+5}$, on each of the stocks' past 20 days' (one month) excess returns, $R_{i,t-\tau}$, where $\tau \in \{0, \dots, 19\}$, and a vector of controls $\mathbf{C}_{i,t}$:

$$R_{i,t+5} = \alpha_t + \sum_{\tau=0}^{19} \beta_{t,\tau} R_{i,t-\tau} + \beta_{t,c} \mathbf{C}_{i,t} + \varepsilon_{i,t}. \quad (1)$$

Here α_t is the intercept in the regression, while $\varepsilon_{i,t}$ is a stock specific error term. As controls in the baseline regression we use two variables that are constructed by multiplying the past month's excess returns with either the logarithm of the stocks' past month's trading volume or the logarithm of the firms' market capitalization at time t . The excess returns are calculated by deducting from stocks' returns the returns to a corresponding equal-weighted Fama-French 48 industry index. The

⁵ We control for trading volume when measuring the returns from providing liquidity. Because of this, we exclude all stocks traded in the Nasdaq as their trading volume is not comparable to that of the NYSE and the Amex traded stocks, due to the differences in trading systems.

approach is similar to that of Hameed and Mian (2012) who also define excess returns relative to industry indexes.^{6,7}

An estimate of the short-term expected excess returns due to return reversals, following day t , is then obtained by combining the moving averages of coefficients $\hat{\beta}_{t,\tau}$ and $\hat{\beta}_{t,C}$ from Equation (1), estimated with data up to period t , with the last 20 days' returns and the values of the controls at time t .

One important consideration in the estimation of any proxy for the returns from providing liquidity is to make an assumption on the investment horizon of the hedge funds' liquidity providing trades, and correspondingly in our setting, to decide whether we should forecast one day, one week, or one month return reversals in a regression such as Equation (1). We do not experiment with other forecasting horizons. In addition, we must decide how the past returns should enter the forecasting regression for returns. Rinne and Suominen (2012) show that the model's explanatory power to forecast one to twenty day short-horizon excess returns improves significantly when they include each of the past twenty days' returns in a forecasting regression instead of just the past month's return as is commonly done. In addition, they show evidence that mean reversion in excess returns is gradual, suggesting that optimal contrarian trading strategies that provide liquidity might

⁶ The two controls are motivated by the findings presented in Campbell et al. (1993), Pastor and Stambaugh (2003), and Khandani and Lo (2011a) that there are cross-sectional differences in the degree of return reversal that are related to size and trading volume.

⁷ We define our excess returns relative to industry indexes as in this case the excess returns for stocks are more likely due only to price pressure from trading and not information. As we show in Section 5, our main results are highly similar if excess returns are calculated relative to an equal-weighted market index as in Lehmann (1990), Lo and MacKinlay (1990), Khandani and Lo (2007, 2011a), and Nagel (2012), or value-weighted CRSP index as in Pastor and Stambaugh (2003). One additional motivation for defining our excess returns relative to industry indexes is that, amongst these alternatives, our method leads to the highest Sharpe ratio for the returns from providing liquidity, suggesting that in practice hedge funds would define excess returns in this context relative to industry indexes.

have longer trading horizon than one day, as is assumed in Khandani and Lo (2011a). In our data, these observations are also valid, motivating the five-day forecast regression specified in Equation (1).

Further motivation for the five-day forecasting period in Equation (1) can be obtained from the results presented in Section 2.4 and the Appendix, where we show that our liquidity providing trading strategy that has a five-day trading horizon has a higher Sharpe ratio than the two alternative liquidity providing trading strategies that we consider, which are similar except that the trading horizon is either one day or one month, and the results presented in Section 3.6, where we show that in practice many hedge funds seem to unwind their liquidity providing reversal trades after five days.

The estimated average coefficients $\hat{\beta}_{t-\tau}$ from Equation (1) are all negative and statistically highly significant, showing that there is a large amount of mean reversion in stocks' excess returns. In addition, the coefficients for the two controls are significant and have the expected sign: the coefficient of volume interacted with past month's return is negative, while the coefficient of market capitalization interacted with past month's return is positive. The regression results are presented in Table I below.

[Insert Table I]

Next, we use these results to estimate the available returns from providing liquidity.

2.3 ESTIMATING THE RETURNS FROM PROVIDING LIQUIDITY

Similarly to Khandani and Lo (2011a), Nagel (2012), and Rinne and Suominen (2011), we proxy for the available returns from providing liquidity by the return to a zero-investment contrarian long-

short trading strategy that utilizes short-term return reversals. More precisely, our proxy for the available returns from providing liquidity is the monthly return to a zero-investment long-short trading strategy where every day a long position is opened in stocks with a positive expected five-day return and a short position in stocks with a negative expected five-day return. After five days, such positions are closed. For reasons discussed in Section 2.2 we focus on a liquidity providing trading strategy where all positions are closed after five days.

Lehmann (1990), Khandani and Lo (2011a), and Nagel (2012) analyze the returns to contrarian trading strategies where portfolios are formed by using the negative of the stocks' past returns as portfolio weights. Given the evidence on return reversal, these portfolio weights effectively correspond to stocks' expected excess returns. In line with their approach, we also use the stocks' expected five-day excess returns evaluated at time t , denoted by $E_t(R_{i,t+5})$, as portfolio weights when forming the long and the short portfolios.⁸

Next, before estimating the returns from providing liquidity, we trim the sample by always excluding one percent of stocks with the highest and the lowest short-term expected returns. We do so mainly as many of the extreme returns seem to be associated with block trades in small capitalization firms with very poor liquidity—trades where hedge funds' ability to provide liquidity can be limited.⁹ In Section 6 we nevertheless include also these omitted stocks and find additional results.

⁸ Besides corresponding with the common portfolio rule in similar contexts in the literature, this approach can be motivated theoretically. Under the assumptions that the short-horizon returns are solely due to price pressure and independent across securities, and the assumption that investors have constant absolute risk aversion utility functions, the investors' optimal portfolio allocations are linear in the expected returns of the assets.

⁹ Avramov et al. (2006) presents related evidence that the largest return reversals occur amongst the most illiquid stocks.

Now, given N stocks in the universe of potential stocks where positions can be taken on day t (the trimmed sample), the assumed portfolio weights $\omega_{i,t}^L$ in the long and $\omega_{i,t}^S$ in the short portfolios on day t are:

$$\begin{aligned}\omega_{i,t}^L &= \frac{I_{E_t(R_{i,t+5}) > 0} E_t(R_{i,t+5})}{\sum_{j=1}^N I_{E_t(R_{j,t+5}) > 0} E_t(R_{j,t+5})} \\ \omega_{i,t}^S &= \frac{-I_{E_t(R_{i,t+5}) < 0} E_t(R_{i,t+5})}{\sum_{j=1}^N I_{E_t(R_{j,t+5}) < 0} E_t(R_{j,t+5})}.\end{aligned}\tag{2}$$

Here I_Z denotes an indicator function that equals one if Z is true and zero otherwise.

When setting the portfolio weights in Equation (2) we assume that the hedge funds' time t estimates of stocks' expected five-day excess returns are based on 120 past days' (i.e., the past 6 months') cross-sectional regressions of Equation (1) up to time $t-6$, the last day for which there is five-day return data at time t . The expected five-day returns at time t , $E_t(R_{i,t+5})$, can then be calculated using the stocks' past twenty days' returns up to time t , past month's trading volume and stocks' market capitalizations at time t .

Our proxy for the returns from providing liquidity, R_{LP} , is the average return on all open positions to the zero-investment long-short trading strategy described above. Table II documents the pre-transaction cost returns on our liquidity providing trading strategy. As is evident from Table II, the returns from providing liquidity are high, even after controlling for standard risk factors.

[Insert Table II]

Figure 1, in turn, shows the time series evolution of the monthly returns from providing liquidity.

[Insert Figure 1]

2.4 COMPARISON TO EXISTING PROXIES

Our measure of the returns from providing liquidity is closely related to several measures of returns to contrarian trading strategies presented in the literature, especially the returns from providing liquidity measure in Nagel (2012) and the returns to the one week contrarian trading strategy introduced in Lehmann (1990). In Lehmann (1990) contrarian portfolios for one week investments are formed on the basis of past week's excess returns. Nagel (2012), in turn, estimates the returns from providing liquidity series as an equal-weighted average return to five long-short portfolios, each of which is formed on the basis of daily excess returns one to five days ago. In both cases either one-day or five-day historical excess returns are thus used to estimate portfolio weights, to engage effectively in five-day trading positions that benefit from return reversal. Hence it is not surprising that the returns to those two contrarian trading strategies are highly correlated with the returns to our liquidity providing trading strategy, as depicted in Table III.

The Lehmann (1990) and the Nagel (2012) measures are, however, not the best proxies of hedge funds' actual realized returns from providing liquidity as they use very limited information, just one-day or five-day excess returns, to set up their reversal trading portfolios. The advantage of our approach is that by using information from twenty individual past daily excess returns in portfolio formation on any given day, and two controls, we are able to construct portfolios that are based on significantly better forecasts of five-day future excess returns, than if we restricted ourselves to use the information contained in the past one-day or five-day excess returns alone. Given these observations, it is hardly surprising that the Sharpe ratio from our investment strategy considerably exceeds those of the contrarian trading strategies presented in Lehmann (1990) and Nagel (2012), as shown in Table III. Because of this, as only public information enters the portfolio

formation in all the approaches, it is more likely that hedge funds follow an investment strategy similar to ours, rather than a strategy that is similar to those presented in Lehmann (1990) and Nagel (2012). Given this, we feel that our proxy of the returns from providing liquidity has significantly better chances to detect hedge funds provision of liquidity from their realized return data as compared to the proxies presented in Lehmann (1990) and Nagel (2012). Table III shows that the Sharpe ratio of our returns from providing liquidity trading strategy exceeds also that of several other contrarian trading strategies, presented in Jegadeesh (1990), Khandani and Lo (2007), and Bali et al. (2011), which also could be thought as proxies of returns from providing liquidity. This is additional supporting evidence for our proxy of the returns from providing liquidity, as the hedge funds most likely want to employ a liquidity providing trading strategy with the highest possible Sharpe ratio.¹⁰

[Insert Table III]

3. Do Hedge Funds Supply or Demand Liquidity?

It is not clear in advance whether hedge funds on average act as market makers and supply liquidity, or demand liquidity in the stock market. While there appears to exist positive returns from providing liquidity, as documented above, many other profitable dynamic trading strategies employed by hedge funds, such as momentum, demand liquidity. In addition, it is not clear whether the returns from providing liquidity that we documented above even exist after accounting for transaction costs (see Avramov et al., 2006). In this section, we explore these issues by regressing the hedge funds' returns on our measure of the returns from providing liquidity. If the mean of the

¹⁰ These conclusions hold if we use the Carhart four-factor excess returns to calculate the Sharpe ratios instead of raw returns.

regression coefficients for all funds is statistically significantly positive, we conclude that hedge funds typically supply liquidity. If the mean coefficient is significantly negative, we conclude that the hedge funds typically demand liquidity. Furthermore, we would regard a significantly positive average coefficient for the returns from providing liquidity as evidence that the returns from providing liquidity are positive even after transaction costs.

3.1 DATA ON HEDGE FUNDS

The data on individual hedge funds are from the TASS database. Our sample includes all 5,800 hedge funds and funds of hedge funds in the TASS database that report their returns in U.S. dollars and have a minimum of 36 monthly return observations between January 1994 and December 2011. Table IV provides the basic summary statistics of variables used in this study.

[Insert Table IV]

3.2 CONTROLLING FOR LIQUIDITY SHOCKS

The analysis regarding whether hedge funds demand or supply liquidity is complicated by the finding in Sadka (2010) that hedge funds have exposure to unexpected shocks to the level of liquidity. As discussed in the introduction, this empirical finding can be understood by looking at some of the strategies that hedge funds employ to benefit from liquidity premium in returns, see e.g., Agarwal et al. (2011) on evidence related to convertible arbitrage. Unexpected liquidity shocks that increase the liquidity premium affect adversely the returns on such investment strategies. As the level of liquidity, the amount of short-term return reversal, and the returns from providing liquidity are all closely related, see e.g., Avramov et al. (2006), we control for liquidity shocks in all the

regressions that examine whether hedge funds supply or demand liquidity in the stock market. As in Sadka (2010), we use unexpected changes in the permanent-variable component of price impact constructed in Sadka (2006) as our measure of liquidity shocks. In Section 5 we show that our results are not affected by our choice of liquidity shock proxy: the results hold if we use the Pastor and Stambaugh (2003) liquidity factor or the change in the VIX volatility index instead of the Sadka (2006) factor.¹¹

Table V shows the correlations between various measures of liquidity and our returns from providing liquidity series, R_{LP} . The correlation between the Sadka liquidity shock series and R_{LP} is -0.31. This is understandable as in more illiquid market the returns from providing liquidity can be expected to be larger. Note also the positive (0.28) correlation of our returns from providing liquidity series with the TED spread, which is defined as the difference between the three-month interbank interest rate and the three-month Treasury Bill rate. Along with the previous literature, see e.g., Brunnermeier et al. (2009), we use this measure as a proxy for the arbitrageurs' funding conditions. Table V indicates that in periods when TED spread is high, the returns from providing liquidity are larger. This finding is consistent with Brunnermeier and Pedersen (2009) who argue that the availability of funding liquidity affects market liquidity.¹²

¹¹ Data for the Sadka (2006) liquidity factors is obtained from Wharton Research Data Services for period 1994-2010. For 2011 we impute the factor values from a regression where the Sadka (2006) factor is regressed on the Pastor and Stambaugh (2003) factor and the change in the VIX volatility index. Our results are unaffected if we use data only until December 2010.

¹² The low correlation of our returns from providing liquidity series with the Acharya and Pedersen (2005) liquidity risk premium series highlights the fact that these returns are compensation for different things. The liquidity risk premium is compensation from holding stocks with high sensitivity to liquidity risk and shorting stocks with low sensitivity to liquidity risk. The returns from providing liquidity, in turn, come from accommodating buying and selling pressure in all kinds of stocks and thus involve frequent buying and shorting of both stocks with high and low sensitivity to liquidity risk.

[Insert Table V]

3.3 HEDGE FUNDS' EXPOSURE TO THE RETURNS FROM PROVIDING LIQUIDITY

We start our empirical analysis by examining whether hedge fund returns are dependent on the returns from providing liquidity. To do so, we regress, fund by fund, the hedge funds' returns on a number of previously used risk factors related to hedge funds, liquidity shocks, as well as the returns from providing liquidity variable, R_{LP} , described earlier. More precisely, we extend the seven hedge fund risk factor model of Fung and Hsieh (2004) by adding the returns from providing liquidity and the unexpected changes in liquidity to the model.¹³ Table VI presents the average loadings of each factor in the model. Table VII presents the cross-sectional means of the coefficients of the returns from providing liquidity, together with the proportion of coefficients that are statistically significant negative or positive at the five-percent level. In addition to the whole sample of 5,800 hedge funds, the results are presented separately for long-short equity funds (1,566 funds), equity market neutral funds (207), event driven funds (430), and other funds (3,597).

[Insert Table VI and VII]

The results presented in Tables VI and VII uniformly support a conclusion that hedge funds, on average, supply liquidity in the stock market. First, the average coefficient of the returns from providing liquidity in the hedge fund return regression is positive and statistically very significant

¹³ The seven factors used by Fung and Hsieh (2004) are three trend following factors for bonds, currencies and commodities (Fung and Hsieh, 2001), an equity market factor (S&P 500), a size spread factor (Russell 2000 minus S&P 500), a bond market factor (Barclays US Aggregate), and a credit spread factor (Barclays US Corporate BAA minus Barclays US Corporate AAA). Data for the trend following factors are available on David Hsieh's website: <http://faculty.fuqua.duke.edu/~dah7/DataLibrary/TF-FAC.xls>.

with a t -statistic of 11.¹⁴ Second, the amount of individual funds that have a statistically significant exposure at a five-percent confidence level to the returns from providing liquidity, i.e. liquidity supplying funds, is 5.9% of all funds. This figure is statistically significantly higher than the threshold figure of 2.5%, which is the percentage of funds that we would expect to find to be statistically significantly positive (or negative) under the assumption that all funds in reality have a zero loading on the returns from providing liquidity. Third, the fraction of significantly negative coefficient funds, i.e. funds that demand liquidity, is only 3.3%.

Different fund categories have different proportions of liquidity suppliers and demanders. The largest proportions of liquidity suppliers are found in the equity market neutral and event driven categories. Both of these categories also have statistically non-significant amount of funds with a negative coefficient to returns from providing liquidity series. The results are reasonable: First, equity market neutral category contains the statistical arbitrage funds, which we expect to engage in this type of liquidity providing trading strategies. Gantchev and Jotikasthira (2013), in turn, show that activist hedge funds, which are usually included in the event driven category, buy stocks that other institutional investors are selling, thus they also are known to provide liquidity. Long-short equity category, in turn, has statistically significant amount of both liquidity suppliers and demanders. It seems that some of these funds employ liquidity providing trading strategies similar to those followed by equity market neutral and event driven funds, whereas other long-short equity hedge funds employ strategies such as momentum trading that demand liquidity. In the other fund categories, there are also statistically significant fractions of both liquidity demanders and suppliers. This category includes for instance funds of funds, convertible arbitrage, and fixed income arbitrage funds. One should note that hedge funds classified in these categories may also operate directly or

¹⁴ Note that we include the Sadka (2006) liquidity shocks in the regressions. Hence the reported exposures to returns from providing liquidity are not driven by the funds' liquidity risk exposure. It seems that these are two separate phenomena as, similar to Sadka (2010), we also find that hedge funds, on average, have a significant positive exposure to the unexpected liquidity shocks beyond their exposures to the returns from providing liquidity.

indirectly in the equity market. For example, we find that convertible arbitrage funds seem to demand liquidity in the equity market when setting up their hedges (for related evidence, see Bechmann, 2004).¹⁵

In the robustness checks section of the paper (Section 5) we investigate, using several alternative measures of the returns from providing liquidity, the robustness of these results. We systematically come to the same conclusion: on average, hedge funds supply liquidity in the stock market. Finally, it should also be noted that our approach of estimating whether hedge funds demand or supply liquidity in the stock market can significantly underestimate the overall role of these funds in the supply of liquidity. This is so because each of the funds may over time act in both roles, as liquidity demanders and as liquidity suppliers. Given such problems in detecting the full extent to which funds are involved in supplying liquidity, our finding that so many funds have systematically statistically significant positive exposure to the returns from supplying liquidity is clear evidence that supply of liquidity is an important strategy for hedge funds.

3.4 SLOW MOVING CAPITAL AND THE FIVE-DAY INVESTMENT HORIZON

We now explore the timing of hedge funds' liquidity providing trades.¹⁶ The evidence in Duffie (2010) suggests that arbitrageurs, such as hedge funds, may need time to release capital from their existing positions and hence may not be able to rapidly react to all emerging trading opportunities. Under such circumstances it would be understandable that reversals occur only gradually. In

¹⁵ It is also possible that the exposure of the other funds category to the returns from providing liquidity is caused by positive correlations in the returns from providing liquidity across markets. For instance, it seems plausible that changes in funding liquidity may cause the returns from providing liquidity in various markets to be positively correlated. Evidence that the returns from providing liquidity could be correlated across markets is provided in Chordia et al. (2005) who show that liquidity in the equity and bond market is positively correlated.

¹⁶ We thank Thierry Foucault for suggesting this idea to us.

addition, lack of liquidity may prevent hedge funds from taking sufficiently large positions during just one or two days that would allow us to statistically significantly detect them from these funds' return series. Given this, it is of interest to study further the timing of hedge funds liquidity providing trades.

Below we probe the timing of hedge funds' exposures to the opportunities to provide liquidity that arrive on any given day t , by studying the future exposure of hedge funds to the same stocks where our returns from providing liquidity trading strategy assumes positions are taken on day t . More precisely, we perform regressions where hedge funds' returns are regressed on the returns to a long-short portfolio that is identical to our time t returns from providing liquidity long-short portfolio, but where the five-day returns are calculated after skipping $n \in \{0, \dots, 10\}$ days. The results are presented in Table VIII.¹⁷

[Insert Table VIII]

The results show, as the percentage of funds exposed to these stocks initially grows, that hedge funds are indeed somewhat slow to react to the emerging trading opportunities, consistent with Duffie (2010). That is, capital moves slowly also in our context. It seems to take one week before hedge funds exposure to the time t returns from providing liquidity portfolio reaches its maximum. Over 17% of all hedge funds in our sample have a statistically significant positive exposure to the return series where we leave four days in between the estimation of the expected reversal and portfolio construction.

It is also interesting to note that consistent with our assumption of a five-day investment horizon, more than 70% of the hedge funds that have a significant exposure to our returns from providing liquidity portfolio formed on day t , no longer have such exposure after five days, that is from day $t+5$ onwards. Only six percent of such funds have any significant exposure to the original

¹⁷ We look at returns during five-day intervals instead of one day returns in order to be able to better identify hedge funds' exposures to this portfolio.

long-short portfolio after ten days. Given these observations, it seems that the five-day investment horizon is appropriate. The fact that hedge funds seem to enter the market only gradually after trading opportunities emerge is additional evidence that the returns from providing liquidity are strictly positive even after transaction costs, in contrast to the arguments presented in Avramov et al. (2006). The last column in Table VIII shows the cumulative percentage of funds that have a statistically significant exposure to the returns from providing liquidity portfolio after skipping $n \in \{0, \dots, 10\}$ days. As the column shows, during the first five days this figure grows as large as 24%.

3.5 FUND CHARACTERISTICS AND THE SUPPLY OF LIQUIDITY

Next we explore the sources of cross-sectional variation in hedge funds' decisions to supply liquidity in the stock market. We approach this question by regressing a dummy variable indicating whether the fund is a liquidity provider on various fund-level characteristics. As it is obvious from Table VIII that some funds enter the liquidity providing trades with a delay, we focus our attention on funds that step into providing liquidity within the first week. Hence, we classify a hedge fund as a liquidity provider if it has a statistically significant positive exposure to any of the liquidity provision strategy returns constructed by leaving 0, 1, 2, 3, and 4 days in between the estimation of expected reversals and the portfolio construction. Altogether, this classification results in 24% of all funds in our sample being flagged as liquidity providers.

As independent variables we use the fund's redemption frequency—defined as the number of redemption opportunities per year—and the length of the lock-up period to measure share restrictions, and the logarithm of average assets under management to measure the fund's size. We also include fund category dummies in each specification and present the results of the probit regressions in Table IX.

[Insert Table IX]

Redemption frequency has a statistically significant negative coefficient, i.e. funds that allow fewer redemptions per year are more likely to be liquidity providers. Also larger funds are more likely to be liquidity suppliers. These findings make sense as low redemption frequency and large fund size both decrease the risk of facing sudden redemptions that are large compared to the funds' assets under management. Thus, the funds with infrequent redemptions and large asset bases can more systematically supply liquidity as compared to the other funds in the market who at times of large redemptions may need to demand liquidity in the market. The coefficient of the lock-up period length has the expected sign but is not statistically significant.

3.6 TIME VARIATION IN HEDGE FUNDS' SUPPLY OF LIQUIDITY

Next, we explore the time variation in hedge funds' decision and ability to supply liquidity. Our approach is to regress, at the fund level, hedge funds' returns on the returns from providing liquidity interacted with several time varying conditioning variables, in order to find if they affect hedge funds' propensity to supply liquidity. As before, we include the Fung and Hsieh (2004) factors and the Sadka (2006) liquidity shocks as controls in the regressions.

Our first conditioning variable is the change in the Treasury-over-Eurodollar spread, the TED spread. Along with e.g., Brunnermeier et al. (2009), we assume that an increase in the TED spread results in higher cost of leverage and thus presumably lower leverage for the hedge funds. Our second conditioning variable is the level of liquidity, which we expect to affect negatively the funds' decisions to supply liquidity. Here we use the Pastor and Stambaugh (2003) measure of stock market liquidity as this is readily available, in contrast to the Sadka (2006) measure, for which

we only have the shocks. The higher the value of the Pastor and Stambaugh (2003) measure of liquidity, the more liquid the U.S. stock market is.

Table X presents the average coefficients of the interaction between returns from providing liquidity and the above mentioned conditioning variables. Figure 2, in turn, shows graphically the relation between these variables and the hedge funds' average propensity to supply or demand liquidity.

[Insert Table X and Figure 2]

As is evident from Table X and Figure 2, hedge funds decrease their supply of liquidity when the TED spread increases, i.e. when the funds' cost of leverage increases. This result is consistent with Brunnermeier and Pedersen (2009), Gromb and Vayanos (2010), and Nagel (2012). Second, as expected, we find that higher stock market liquidity leads hedge funds to decrease their supply of liquidity in the stock market. This presumably happens as the available returns from supply of liquidity decrease at the same time as the costs of liquidity in other liquidity demanding trading strategies decrease.¹⁸

As an alternative way of probing the time variation in hedge funds' liquidity supply and demand, we run rolling regressions using past three years of data and report the fractions of statistically significant liquidity suppliers and demanders in Figure 3. One interesting finding in Figure 3 is that there are two sudden increases in the number of liquidity demanding hedge funds that quickly dissipate. The two spikes in the demand for liquidity coincide with the quant crisis in August 2007 and the bankruptcy of derivatives broker MF Global in 2011.

¹⁸ We also use hedge fund flows as a conditioning variable, but these results were not statistically significant and are hence not reported.

[Insert Figure 3]

4. Hedge Funds and Stock Market Liquidity

4.1 OVERVIEW

In this section we study the effect of hedge funds on short-term return reversals, liquidity, and volatility. We are especially interested in the effects of two measures of hedge fund industry capital on these three variables of interest. These results help us evaluate further the role that hedge funds play in the supply of liquidity in the stock market.

Our first proxy for changes in the hedge fund industry speculative capital is the flow of funds to the hedge funds industry scaled by the U.S. stock market capitalization.¹⁹ We use the flow, rather than change in assets under management (AUM), because of the endogeneity problem of the latter. We believe that using the fund flows is relatively free of endogeneity problems as hedge funds typically require a notification period of up to several months prior to subscriptions and redemptions.

Second, we need a proxy for the changes in the total funding capital that hedge fund industry employs. To build such a proxy, we first use the difference between the three-month Eurodollar and the three-month Treasury interest rates, i.e. TED spread, as a measure of hedge funds' funding cost. Higher TED spread indicates higher cost of funding and lower leverage. The total amount borrowed by hedge funds depends not only on the cost of leverage but also on the level of hedge funds' equity. Hence, we multiply the TED spread by the hedge fund industry AUM normalized by the market capitalization of the U.S. equity market to obtain a measure of the hedge fund industry's total cost of leverage, a variable that inversely proxies for the total amount borrowed by hedge funds (relative to the U.S. equity market capitalization). We use the changes in the hedge fund

¹⁹ Our estimates for hedge fund industry flows are constructed similarly as in Jylhä and Suominen (2011).

industry's total cost of leverage as our proxy for the changes in total borrowed speculative capital, and as our second proxy for changes in the hedge fund industry speculative capital.

4.2 HEDGE FUNDS AND RETURN REVERSALS

An increase in the amount of capital allocated to hedge funds that provide liquidity should decrease the returns available from such activity. In our context, this means that we should expect that hedge fund flows reduce the degree of short-term return reversals.

The inclusion of controls in regression of Equation (1) improves our forecasts of short-term returns, but makes inference on the amount of return reversal difficult. Hence, to obtain an estimate of the degree of return reversal, we first repeat the daily regressions of Equation (1) but now without controls. The time series of the negative of the coefficient for the previous day's return, $-\beta_{t,0}$, is then a time-varying measure of the amount of short-term return reversal. The time series for $-\beta_{t,0}$ is presented in Figure 4.

[Figure 4]

To study the effect of hedge funds on short-term return reversals, we now regress the monthly averages of the $-\beta_{t,0}$ estimates on our two measures of changes in hedge fund industry capital. The results are presented in Table XI.

[Insert Table XI]

These results show that hedge fund flows reduce the amount of short-term return reversal as expected. The effect of hedge fund leverage is statistically insignificant, however.

4.3 HEDGE FUNDS AND LIQUIDITY

In Table XII we report the results of regressing changes in the level of liquidity on hedge fund flows and changes in the cost of leverage. We use two measures of liquidity in this regression. First, we measure liquidity shocks by the unexpected changes in the Sadka (2006) permanent-variable price impact. Second, we look at the liquidity shocks as defined in Pastor and Stambaugh (2003). In both cases, a positive coefficient for hedge fund flows would indicate that hedge fund flows increase liquidity.

[Insert Table XII]

The results presented in Table XII show that hedge fund flows are associated with improvements in liquidity. This means that speculative capital affects liquidity in the stock market, as predicted e.g., by Gromb and Vayanos (2010), and shows that hedge funds play a significant role in the supply of liquidity in the stock market. Also, changes in the cost of leverage are negatively associated with market liquidity, consistent with our expectation.

4.4 HEDGE FUNDS AND VOLATILITY

Several papers, see e.g., Brunnermeier and Pedersen (2009), link volatility with the level of liquidity. The idea is that in illiquid markets investors' order imbalances cause additional short-term price volatility. Naturally, volatility is lower when arbitrageurs have abundant capital to supply liquidity in the stock market.

In Table XIII we report the results of regressing the monthly changes in the average stock return volatility of the NYSE and the Amex stocks on our proxies for changes in speculative

capital. Given the role of hedge funds in the supply of liquidity, we expect that hedge fund flows have a negative effect and the changes in cost of leverage have a positive effect on volatility.

[Insert Table XIII]

The effect of hedge fund flows lacks statistical significance. Instead, the effect of the change in the cost of leverage is positive, as predicted, and statistically highly significant. This result supports the idea that the availability of speculative capital, more precisely the availability of funding capital to hedge funds affects the stock market volatility, as argued by Brunnermeier and Pedersen (2009). These findings are consistent with the findings in Kang et al. (2012).

5. Alternative Measures of the Returns from Providing Liquidity

In this section we show that our main result, that hedge funds, on average, provide liquidity in the stock market, is robust to alternative methods for estimating the returns from providing liquidity and liquidity shocks. In addition to testing the robustness of our results with respect to skipping n days before taking positions, as is done in Table XII, and the inclusion of extreme returns that is considered in Section 6, we now consider the following additional changes to our approach of estimating the returns from providing liquidity:

1. We consider using the value-weighted and equal-weighted CRSP indexes, instead of the industry indexes, as benchmarks when calculating stocks' excess returns.
2. We consider replacing the Sadka (2006) liquidity shock measure with alternative proxies: either the Pastor and Stambaugh (2003) liquidity factor or the change in the VIX volatility index.

Table XIV presents the equivalent of the entire sample result in Table VI for these alternative measures of the returns from providing liquidity and liquidity shocks. As Table XIV shows, our main result that hedge funds supply liquidity is robust to these alternative specifications.

[Insert Table XIV]

6. Extreme Returns and Hedge Funds' Demand of Liquidity

In estimating the returns from providing liquidity we trim the sample by excluding one percent of stocks with the highest and lowest short-term expected return. In this section, we study separately hedge funds' exposure to the returns from providing liquidity within these extreme (expected) return stocks.

Above we exclude the 2% of extreme return stocks on the grounds that these stocks are on average significantly smaller and less liquid based on the Amihud (2002) measure of illiquidity than other stocks in our sample. Therefore many of these stocks are such where it is difficult for hedge funds to provide liquidity. Similarly, many of these stocks are such where hedge funds' demand of liquidity should not materially affect their portfolio returns. In the latter part of the sample these observations are, however, no longer always valid, as many of the extreme return stocks are then relatively large capitalization stocks. In line with this, we document below significant effects on hedge funds' returns from their supply and demand of liquidity in the extreme return stocks.

To analyze hedge funds' exposure to the extreme return stocks, we augment our baseline fund-level regression by adding the returns from providing liquidity to the excluded stocks as an additional factor along with the seven Fung and Hsieh (2004) factors, the Sadka (2006) liquidity factor, and our former measure of the returns from providing liquidity, R_{LP} . The returns from providing liquidity series for the extreme return stocks, is constructed similarly as in Section 3, but

in this case we only use the stocks that were earlier excluded from the investment universe, i.e. the top and bottom expected return percentiles. The aggregate results are reported in Table XV.

[Insert Table XV]

It seems that hedge funds, on aggregate, are exposed negatively to the returns from providing liquidity on the extreme return stocks. The fraction of significantly negatively exposed funds is 14.9%, while the fraction of funds with significantly positive exposure is only 2.7%. This indicates that hedge funds more commonly act as liquidity demanders than suppliers in these stocks. Note that inclusion of this variable does not affect our earlier result that hedge funds supply liquidity for the trimmed sample of stocks.

There is one reason why hedge funds can be expected to demand liquidity in the extreme return stocks: The mere fact that hedge funds, who typically supply liquidity, demand liquidity in some stocks, implies that there are less agents that supply liquidity in these stocks. Given this, there are fewer institutions in the market compared to other times who moderate the price effects that originate from demand for liquidity. Under these circumstances, the hedge funds demand for liquidity may in itself be the cause of the extreme returns; and it is because of this that hedge funds seem to demand liquidity in the extreme return stocks.

To investigate hedge funds' demand for liquidity in extreme stocks further, we create two separate series of returns from providing liquidity to extreme return stocks: one series for financial stocks alone and one for non-financials.²⁰ We then regress hedge fund returns on these two new measures of returns from providing liquidity in addition to R_{LP} and the same controls as previously.

²⁰ Banking, Insurance, and Trading are classified as financial stocks of the 48 Fama-French industries. To keep the portfolio size large enough, we now include in case of the financial stocks the top and bottom expected return vigintiles in our extreme expected return long-short portfolio.

The results are reported in the lower panel of Table XV. Interestingly, it turns out that there is a large difference in the results between these two groups of extreme return stocks. Hedge funds have only a small negative exposure to the returns from providing liquidity to the extreme return non-financial stocks but a large negative exposure to that for the extreme return financial stocks. This shows that the hedge funds' demand for liquidity in the extreme return stocks is mainly due to their demand for liquidity in the financial stocks.

Finally, we plot the time-varying proportion of liquidity suppliers and demanders in the extreme return stocks in Figure 5. Again, as Figure 3, this figure is based on rolling fund-level regressions using at each point the previous three years of data.

[Insert Figure 5]

Two main observations emerge from these results: First, combining the evidence in Figures 3 and 5, it seems that while hedge funds typically supply liquidity, they can significantly demand liquidity in crisis periods—the 2007 quant crisis, the 2008 financial crisis, and the 2011 collapse of MF Global. This finding is consistent with the findings in Aragon and Strahan (2011), Ben-David et al. (2012), and Cella et al. (2013). During the financial crisis the hedge funds' demand for liquidity is highest in the extreme return stocks; given the results in Table XV, it thus seems that hedge funds demand liquidity in the extreme return stocks mainly because of their demand for liquidity in financial stocks during the financial crisis. The second observation from Figure 5 is that hedge funds' supply of liquidity in extreme reversal stocks was very high prior to the quant crisis: As many as 35% of all hedge funds were supplying liquidity on these stocks at that time.

7. Changes in the Market Structure and the Supply of Liquidity

In this section we discuss evidence that the introduction of automated quote dissemination (autoquote), which fosters high frequency trading, at the NYSE has affected both the selection of stocks where hedge funds supply liquidity, and the speed at which hedge funds enter and exit their reversal trades. First note from Figure 3 and Figure 5 that after the 2003 introduction of autoquote, hedge funds started to reduce their supply of liquidity in the trimmed sample of stocks and increase their supply of liquidity in the extreme return stocks. Table XVI, in turn, shows that prior to autoquote, as much as 11% of hedge funds had statistically significant positive exposure to our returns from providing liquidity portfolio, while under autoquote the corresponding figure is only 5%. Nonetheless, if we again look at the hedge funds exposure to the returns from providing liquidity portfolio after one to ten days, as shown in Table XVI, it is clear that the provision of liquidity has only become more common amongst hedge funds over time. The results in Table XVI thus suggest that the presence of high frequency traders has altered the speed at which hedge funds provide liquidity. It seems that after the introduction of autoquote at the NYSE hedge funds both enter and exit later from their reversal trades.

[Insert Table XVI]

8. Conclusion

In this paper we study whether hedge funds demand or supply liquidity in the stock market by regressing hedge fund returns on the returns to a contrarian trading strategy. Our main finding is that hedge funds typically supply liquidity in the stock market.

We find also that large funds and funds that offer investors less frequent redemption opportunities have a higher propensity to supply liquidity in the stock market compared to other hedge funds. We find two time-varying factors that affect hedge funds' propensity to supply liquidity. First, hedge funds are more likely to supply liquidity when liquidity is low and, second, when credit market is easing. Although the hedge funds typically supply liquidity in the stock market, we find evidence that hedge funds often demand liquidity during crises, and largely as a result of this the most extreme cross-sectional returns are associated with stocks where hedge funds, at that point in time, demand liquidity.

We find support for the theories set forth in Brunnermeier and Pedersen (2009) and Gromb and Vayanos (2010) that link market liquidity with the amount of speculative capital and funding liquidity to speculators. In particular, we show that the amount of speculative capital—either hedge fund flows or the amount of funding credit—affects the level of liquidity, the amount of short-term return reversal, and stock return volatility.

Appendix

The Effect of Holding Period on the Returns from Providing Liquidity

Table AI. Return statistics for liquidity providing trading strategies with broker commissions

Monthly return statistics for liquidity providing trading strategies with different holding periods (1, 5 and 20 days) are calculated for the time period 8/1990-12/2011. We use the following estimates for broker commissions: 5 basis points per trade during the 1990's and 3 basis points per trade after 2000. These estimates are based on de Groot et al. (2012). All returns from providing liquidity series are returns to a zero-investment long-short trading strategy similar to that described in Section 2, in which expected holding period excess returns are used as portfolio weights when forming the long and the short portfolios (similarly as in Equation (2)). The expected returns are calculated using six-month moving averages of coefficients for return reversal, from regressions similar to Equation (1), but with stocks' one day, five day or 20-day future excess return as dependent variables, until two, six or 21 days prior to taking positions. Return statistics are based on averages of the returns of all open positions.

	1 day	5 days	20 days
Mean	-2.22	0.52	0.15
25th percentile	-3.86	-0.47	-0.41
Median	-2.11	0.43	0.22
75th percentile	-0.82	1.41	0.78
Volatility	3.03	1.83	1.36
Positive return %	17.4%	61.3%	61.7%
Sharpe ratio	-0.73	0.29	0.11

As Table AI shows, the Sharpe ratio is the highest assuming a five-day holding period.

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Table I. Pattern of return reversal

This table shows the average coefficients, $\hat{\beta}_{t-\tau}$, from daily cross-sectional regressions (Equation (1) in the text) where stocks' five-day future excess returns R_{t+5} are regressed on each of the stocks' past twenty days' excess returns, $R_{t-\tau}$, where $\tau \in \{0, 1, 2, 3, \dots, 19\}$ and controls $\ln(\text{Volume}) \times R_{t,t-19}$ and $\ln(\text{Market Capitalization}) \times R_{t,t-19}$. Here $R_{t,t-19}$ refers to past 20 days' excess return. The excess returns are calculated relative to the corresponding equal-weighted Fama-French 48 industry index returns. Sample period is from 1/1990 through 12/2011. T-statistics based on Fama-MacBeth standard errors adjusted for first order autocorrelation are shown next to coefficients in the parentheses. All coefficients that are statistically significant at the 5% level are bolded.

R_{t+5}					
R_t	-0.176	(-39.07)	R_{t-10}	-0.047	(-11.11)
R_{t-1}	-0.109	(-24.50)	R_{t-11}	-0.045	(-10.78)
R_{t-2}	-0.089	(-20.64)	R_{t-12}	-0.043	(-10.54)
R_{t-3}	-0.077	(-17.92)	R_{t-13}	-0.045	(-11.12)
R_{t-4}	-0.068	(-16.05)	R_{t-14}	-0.044	(-10.73)
R_{t-5}	-0.063	(-14.48)	R_{t-15}	-0.043	(-10.22)
R_{t-6}	-0.058	(-14.20)	R_{t-16}	-0.042	(-9.82)
R_{t-7}	-0.056	(-13.71)	R_{t-17}	-0.040	(-10.04)
R_{t-8}	-0.051	(-12.53)	R_{t-18}	-0.039	(-10.01)
R_{t-9}	-0.048	(-11.53)	R_{t-19}	-0.037	(-10.30)
Intercept				-0.001	(-10.66)
Controls:					
$\ln(\text{Volume}) \times R_{t,t-19}$				-0.002	(-2.08)
$\ln(\text{Market Capitalization}) \times R_{t,t-19}$				0.006	(4.73)
Number of daily regressions				5523	
Average number of observations				1970	
Average Adjusted R^2				0.067	

Table II. Return statistics for the liquidity providing trading strategy

This table shows the statistics of daily and monthly returns from providing liquidity. Sample period is from 8/1990 through 12/2011. These are returns on a zero-investment long-short trading strategy in which five-day expected excess returns are used as portfolio weights when forming the long and the short portfolios for five-day investments (as in Equation (2)). When constructing the portfolio, the expected returns are calculated using six month moving averages of coefficients for return reversal (as in Equation (1)) until six days prior to taking positions. Return statistics are based on averages of the returns of all open positions. There is no consideration for transaction costs. Carhart 4-factor alpha is calculated using data from Kenneth French's website.

	Daily	Monthly
Mean	0.06	1.18
25th percentile	-0.10	0.15
Median	0.05	1.03
75th percentile	0.20	2.15
Volatility	0.35	1.89
Positive return %	59.4%	78.7%
Sharpe ratio	0.16	0.63
4-factor alpha	0.04	0.98
t-statistics for alpha	(8.19)	(6.10)

Table III. Alternative measures of returns from providing liquidity

This table shows the monthly return statistics for several contrarian trading strategies that can be thought of as returns from providing liquidity. Sample period is from 1/1994 through 12/2011. *Jegadeesh (1990)* is the equal-weighted return difference of top and bottom deciles based on the previous month's returns. *Lehmann (1990)* is the return to a long-short trading strategy that uses previous week's excess returns as portfolio weights and has a one week holding period. *Khandani and Lo (2007)* is the return to a long-short trading strategy that uses previous day's excess returns as portfolio weights and has a one day holding period. *Bali et al. (2011)* is the value weighted return difference of top and bottom deciles based on the previous month's maximum daily returns. *Nagel (2012)* is the average return of five overlapping long-short portfolios that use the previous day's excess returns as portfolio weights and have effectively a five day holding period (with daily rebalancing). Panel A shows the returns statistics to these strategies calculated using the same data requirements as in the original articles. Panel B on the other hand shows the returns statistics to all the strategies using only the NYSE and Amex traded stocks and imposing the data restrictions presented in Section 2.1—i.e. excluding the smallest size decile, stocks with price less than \$5 and stocks without any trading on the day when position is presumably opened. Panel C shows the correlations of monthly returns calculated using the Panel B data requirements.

	Jegadeesh (1990)	Lehmann (1990)	Khandani and Lo (2007)	Bali et al. (2011)	Nagel (2012)	R_{LP}
A: Series replicated using the same data requirements as in the original articles						
Mean	2.33	2.61	15.25	0.79	8.16	
Volatility	9.28	4.64	14.22	10.43	5.19	
Sharpe ratio	0.25	0.56	1.07	0.08	1.57	
B: Series replicated using the same data requirements as in the R_{LP} calculations						
Mean	0.89	1.12	-0.28	0.62	0.71	0.94
Volatility	5.00	3.73	4.28	6.49	1.85	1.87
Sharpe ratio	0.18	0.30	-0.06	0.10	0.38	0.50
C: Correlations between alternative liquidity providing trading strategies						
	Jegadeesh (1990)	Lehmann (1990)	Khandani and Lo (2007)	Bali et al. (2011)	Nagel (2012)	
Lehmann (1990)	0.34					
Khandani and Lo (2007)	0.15	-0.01				
Bali et al. (2011)	-0.13	0.07	0.04			
Nagel (2012)	0.34	0.48	0.44	0.05		
R_{LP}	-0.08	0.41	-0.06	0.18	0.40	

Table IV. Descriptive statistics

This table presents the descriptive statistics of the variables used in the paper. The sample period is from 1/1994 through 12/2011. All data are on a monthly frequency.

	Mean	St.dev.	25%	Median	75%
<u>Hedge fund specifics</u>					
Return	0.47%	4.54%	-0.97%	0.45%	1.88%
Redemption frequency, per year	14.9	45.4	4	12	12
Lock-up, days	95.3	194.1	0.0	0.0	30.0
AUM average, million USD	141.2	281.6	19.9	51.3	145.4
<u>Time series</u>					
Return from providing liquidity	0.94%	1.87%	-0.12%	0.77%	1.95%
Liquidity shock (Sadka, 2006) ¹	0.0004	0.0060	-0.0024	0.0011	0.0035
Liquidity shock (Pastor and Stambaugh, 2003)	0.0005	0.0682	-0.0301	0.0034	0.0365
Liquidity level (Pastor and Stambaugh, 2003)	-0.0305	0.0750	-0.0599	-0.0231	0.0163
Hedge fund flow (% of AUM)	0.59%	1.79%	-0.23%	0.74%	1.64%
Hedge fund flow (\$, scaled by U.S. equity market cap) ²	0.1446	1.8377	-0.0977	0.4156	0.8823
Change in TED spread	0.00%	0.36%	-0.10%	-0.01%	0.07%
Change in cost of leverage ²	-0.0028	0.4254	-0.0455	-0.0035	0.0409
Return reversal	0.120	0.062	0.082	0.119	0.162
Change in stock market volatility	0.00%	0.50%	-0.19%	-0.02%	0.13%

¹ For 2011 we use imputed values of the Sadka (2006) liquidity shock, as described in Footnote 10.

² Figures for hedge fund flow (\$, scaled by U.S. equity market cap) and change in cost of leverage multiplied by 1000.

Table V. Correlation of R_{LP} with liquidity factors

This table shows the correlations of our returns from providing liquidity series, R_{LP} , with several liquidity factors using monthly data. Sample period is from 1/1994 through 12/2011, except for Sadka (2006) for which we have data only until 12/2010. *Acharya and Pedersen (2005)* is the liquidity risk premium. *Pastor and Stambaugh (2003)* is the liquidity innovation series, downloaded from Lubos Pastor's webpage. *Sadka (2006)* is the shock to the permanent variable component of price impact, downloaded from Ronnie Sadka's webpage. *TED spread* is the difference between the three-month Eurodollar and the three-month Treasury interest rates. *VIX* is the monthly average of daily VIX index values.

	Acharya and Pedersen (2005)	Pastor and Stambaugh (2003)	Sadka (2006)	TED spread	VIX
Pastor and Stambaugh (2003)	0.03				
Sadka (2006)	0.16	0.20			
TED spread	-0.17	-0.24	-0.40		
VIX	-0.17	-0.23	-0.20	0.46	
R_{LP}	-0.25	0.02	-0.30	0.28	0.16

Table VI. Hedge funds' exposure to the returns from providing liquidity, liquidity shocks, and common risk factors

This table presents average loadings of individual hedge funds on our measure of the returns from providing liquidity, R_{LP} , the Sadka (2006) liquidity shock, and the seven Fung and Hsieh (2004) risk factors. Figures significant at a 5% level are bolded. There are a total of 5,800 hedge funds in the sample.

	(1)	(2)
Intercept	0.293	0.256
	(33.68)	(27.83)
R_{LP}		0.047
		(10.98)
Liquidity shock	0.396	0.425
	(26.52)	(28.27)
<u>Fung and Hsieh risk factors</u>		
Bond trend	-0.005	-0.006
	(-9.73)	(-10.50)
Currency trend	0.007	0.007
	(18.17)	(17.83)
Commodity trend	0.013	0.013
	(21.87)	(22.71)
Equity	0.240	0.240
	(51.78)	(51.69)
Size	0.075	0.078
	(21.39)	(22.19)
Bond	0.026	0.031
	(3.70)	(4.41)
Credit	0.327	0.342
	(21.42)	(21.83)

Table VII. Hedge funds' exposure to the returns from providing liquidity by hedge fund categories

This table presents summary statistics of the coefficients for the returns from providing liquidity, R_{LP} , from regressions of hedge fund returns on R_{LP} , the seven Fung and Hsieh (2004) risk factors, and the Sadka (2006) liquidity shock by hedge fund categories. *Mean* gives the average of the coefficients for each of the hedge fund categories, *Negative* (*Positive*) gives the proportion of coefficients that are significantly negative (positive) at a 5% level using autocorrelation and heteroscedasticity consistent standard errors (Newey and West, 1987). For means, the *t*-statistics for the test of zero mean are given in the parentheses. For the proportion of negatives and positives, the figures in the parentheses are z-statistics testing whether the proportion is equal to 2.5%, which it would be under no exposure to provision of liquidity. Figures significant at a 5% level are bolded. There are a total of 5,800 hedge funds in the sample out of which 1,566 are classified as long-short equity funds, 207 as equity market neutral funds, 430 as event driven funds, and 3,597 as other funds.

	Mean	Negative	Positive
All funds	0.047 (10.98)	3.3% (3.87)	5.9% (16.74)
Long-short equity	0.064 (6.44)	4.0% (3.86)	6.8% (10.82)
Equity market neutral	0.047 (3.26)	1.9% (-0.52)	7.7% (4.82)
Event driven	0.100 (8.26)	1.6% (-1.16)	10.7% (10.89)
Other funds	0.033 (6.51)	3.3% (2.89)	4.9% (9.19)

Table VIII. Timing of hedge funds' liquidity providing trades

This table presents the summary statistics of the coefficients from regressions where hedge funds' monthly returns are regressed on the returns to long-short portfolios that make five-day investments identical to those in our time t returns from providing liquidity portfolio, but where the investments are made after skipping n days. In this way, we explore the timing of the hedge funds' supply of liquidity. In these regressions, we use as controls the seven Fung and Hsieh (2004) risk factors and the Sadka (2006) liquidity shock. *Mean* gives the average of the coefficients for all funds, *Negative* (*Positive*) gives the proportion of coefficients that are significantly negative (positive) at a 5% level using autocorrelation and heteroscedasticity consistent standard errors (Newey and West, 1987). For means, the t -statistics for the test of zero mean are given in the parentheses. For the proportion of negatives and positives, the figures in the parentheses are z -statistics testing whether the proportion is equal to 2.5%, which it would be under no exposure to provision of liquidity. Figures significant at a 5% level are bolded. *Remaining* gives the percentage of those funds that on day t statistically significantly demand or provide liquidity (3.3% and 5.9% of all funds, respectively), which also have a statistically significant exposure at a 5% level to that same long-short portfolio in the five days that follow after skipping n days. *Cumulative* shows the cumulative percentage of funds that have a statistically significant exposure to at least one of the series formed by skipping $j \leq n$ days.

	Mean	Negative	Positive	Negative		Positive	
				Remaining	Cumulative	Remaining	Cumulative
0 days	0.047 (10.98)	3.3% (3.87)	5.9% (16.74)	100.0%	3.3%	100.0%	5.9%
1 day	0.069 (16.14)	2.7% (0.76)	8.1% (27.17)	59.2%	4.0%	77.0%	9.4%
2 days	0.082 (19.33)	2.5% (0.00)	11.6% (44.57)	35.6%	4.8%	63.7%	14.4%
3 days	0.088 (20.73)	2.8% (1.60)	15.4% (63.08)	21.5%	5.8%	51.2%	20.1%
4 days	0.096 (22.81)	3.2% (3.53)	17.1% (71.40)	12.0%	6.8%	37.8%	24.1%
5 days	0.094 (21.86)	2.9% (1.85)	16.7% (69.13)	9.9%	7.1%	29.7%	26.3%
6 days	0.053 (12.42)	3.4% (4.46)	9.6% (34.48)	4.7%	8.1%	18.3%	27.2%
7 days	0.007 (1.89)	3.7% (6.06)	4.8% (11.10)	2.1%	9.4%	8.7%	27.8%
8 days	-0.020 (-5.49)	3.8% (6.14)	3.2% (3.28)	2.6%	10.4%	7.3%	28.3%
9 days	-0.022 (-5.99)	4.2% (8.33)	3.4% (4.21)	3.1%	11.6%	8.4%	28.9%
10 days	-0.038 (-9.95)	5.9% (16.65)	2.9% (1.93)	3.1%	13.8%	6.1%	29.4%

Table IX. The effect of fund characteristics on the provision of liquidity

This table presents the results of probit regressions of a liquidity provider dummy on fund characteristics. The dependent variable is equal to one if the fund is classified as a liquidity provider and zero otherwise. The explanatory variables are as follows: *Redemption frequency* is the number of redemptions allowed by the fund per year, *Lock-up period* is the length of the lock-up period in years, and *Size* is the logarithm of the average assets under management of the fund over the sample period. Fund style dummies are included in all specifications. As it is clear from Section 3.4 that hedge funds enter into liquidity provision only gradually, we construct the liquidity provider dummy as follows: As in Table VIII, we first regress hedge funds' monthly returns on the returns to long-short portfolios that make five-day investments identical to those in our time t returns from providing liquidity portfolio, but where the investments are made after skipping 0, 1, 2, 3, and 4 days. In these regressions we control for the Fung and Hsieh (2004) risk factors and the Sadka (2006) liquidity shock. We then classify a fund as a liquidity provider if it has a statistically significant positive exposure to the returns from providing liquidity series in any of these five regressions. In the probit regression below, heteroscedasticity consistent t-statistics are reported in the parentheses. Figures significant at a 5% level are bolded.

	(1)	(2)	(3)	(4)
Redemption frequency	-0.001 (-1.90)			-0.002 (-2.31)
Lock-up period		0.012 (0.35)		0.030 (0.75)
Size			0.041 (2.62)	0.043 (2.69)
Pseudo-R ²	0.027	0.026	0.028	0.030
Observations	5,512	5,800	3,961	3,817

Table X. Time variation in hedge funds' supply of liquidity

This table presents average coefficients for the returns from providing liquidity (R_{LP}) and the interaction of R_{LP} and conditioning variables in a regression of hedge fund returns on the seven Fung and Hsieh (2004) risk factors, R_{LP} , Sadka (2006) liquidity shocks, and the conditioning variables. The conditioning variables are the one-month *lagged liquidity level* (Pastor and Stambaugh, 2003) and the *change in TED spread* (Treasury-over-Eurodollar), which proxies for the cost of leverage. The *t*-statistics for the test of zero mean are given in the parenthesis. Figures significant at a 5% level are bolded. There are a total of 5,800 hedge funds in the sample.

	(1)	(2)	(3)
R_{LP}	0.025 (5.71)	0.057 (12.83)	0.048 (10.72)
$R_{LP} \times$ Lagged liquidity level	-0.917 (-14.17)		-0.605 (-9.33)
$R_{LP} \times$ Change in TED spread		-9.763 (-4.70)	-12.410 (-5.84)

Table XI. Speculative capital and return reversals

This table presents the results of regressing the monthly averages of our five-day return reversal estimates on measures of changes in speculative capital. Our five-day return reversal estimate is the negative of the coefficient for the previous day's return, $-\hat{\beta}_t$, from daily regressions where stocks' five-day future excess returns are regressed on each of the stocks' past 20 days' excess returns. Excess returns are calculated relative to the corresponding equal-weighted Fama-French industry indexes. *Hedge fund flow* is the net flow of capital into the hedge fund industry divided by the market capitalization of the U.S. equity market. *Change in cost of leverage* is calculated as the change in the difference between the three month Eurodollar and the three month Treasury interest rates, multiplied by the lagged hedge fund AUM relative to the U.S. stock market capitalization. The sample period is from 1/1994 to 12/2011. Autocorrelation and heteroscedasticity consistent *t*-statistics (Newey and West, 1987) are reported in the parentheses. Coefficients significant at a 5% level are bolded.

Intercept	0.122 (18.38)
Hedge fund flow	-6.940 (-2.38)
Change in cost of leverage	-0.094 (-0.01)
Adjusted R^2	0.032
Monthly observations	216

Table XII. Speculative capital and liquidity

This table presents the results from regressing monthly innovations in liquidity on measures of changes in speculative capital. As measures of liquidity, we use the permanent-variable component of price impact defined in *Sadka* (2006) and the *Pastor and Stambaugh* (2003) liquidity factor. *Hedge fund flow* is the net flow of capital into the hedge fund industry divided by the market capitalization of the U.S. equity market. *Change in cost of leverage* is calculated as the change in the difference between the three month Eurodollar and the three month Treasury interest rates, multiplied by the lagged hedge fund AUM relative to the U.S. stock market capitalization. The sample period is from 1/1994 to 12/2010 (*Sadka*) or to 12/2011 (*Pastor and Stambaugh*). Autocorrelation and heteroscedasticity consistent *t*-statistics (Newey and West, 1987) are reported in the parentheses. Coefficients significant at a 5% level are bolded.

	Sadka (2006)	Pastor and Stambaugh (2003)
Intercept	0.000 (0.19)	-0.002 (-0.46)
Hedge fund flow	1.242 (5.10)	9.786 (3.30)
Change in cost of leverage	-5.689 (-7.84)	-38.347 (-5.67)
Adjusted R^2	0.216	0.083
Monthly observations	204	216

Table XIII. Speculative capital and volatility

This table presents the results of regressing the monthly change in stock return volatility on measures of changes in speculative capital. *Volatility* is measured as an equal-weighted average of standard deviations of individual stocks' daily returns. *Hedge fund flow* is the net flow of capital into the hedge fund industry scaled by the market capitalization of the U.S. equity market. *Change in cost of leverage* is the change in the difference between the three month Eurodollar and the three month Treasury interest rates, multiplied by the lagged hedge fund AUM relative to the U.S. stock market capitalization. The sample period is from 1/1994 to 12/2011. Autocorrelation and heteroscedasticity consistent *t*-statistics (Newey and West, 1987) are reported in the parentheses. Coefficients significant at a 5% level are bolded.

Intercept	-0.000 (-0.13)
Hedge fund flow	0.154 (0.42)
Change in cost of leverage	2.983 (3.88)
Adjusted R^2	0.066
Monthly observations	216

Table XIV. Robustness checks

This table shows the whole sample results of Table VII using alternative definitions of excess return and liquidity shock. Instead of defining the excess returns relative to the Fama-French industry indexes, we consider defining them relative to the CRSP equal-weighted and value-weighted indexes. As alternative measures of liquidity shocks we consider the Pastor and Stambaugh (2003) liquidity shock and the change in the VIX index. Mean gives the average of the coefficients for all hedge funds, and *Negative (Positive)* gives the proportion of coefficients that are significantly negative (positive) at a 5% level using autocorrelation and heteroscedasticity consistent standard errors (Newey and West, 1987). For means, the *t*-statistics for the test of zero mean are given in the parentheses. For the proportion of negatives and positives, the figures in the parentheses are *z*-statistics testing whether the proportion is equal to 2.5%, which it would be under no exposure to provision of liquidity. Figures significant at a 5% level are bolded. There are a total of 5,800 hedge funds in the sample.

	Average	Negative	Positive
Baseline result	0.047 (10.98)	3.3% (3.87)	5.9% (16.74)
<u>Alternative definitions of excess returns</u>			
EW CRSP	0.044 (15.23)	2.8% (1.26)	6.9% (21.28)
VW CRSP	0.046 (15.33)	2.9% (1.77)	7.6% (24.73)
<u>Alternative liquidity shock measures</u>			
Pastor and Stambaugh (2003)	0.025 (5.90)	4.1% (7.57)	5.1% (12.78)
Change in VIX	0.015 (3.45)	4.6% (10.09)	5.0% (12.36)

Table XV. Hedge funds' exposure to the extreme return reversals

This table presents summary statistics of the coefficients of the returns from providing liquidity to stocks with extreme returns, from regressions of hedge fund returns on the seven Fung and Hsieh (2004) risk factors, returns from providing liquidity to the trimmed sample of stocks, returns from providing liquidity to stocks with extreme returns, and the Sadka (2006) liquidity shock. *Mean* gives the average of the coefficients, *Negative (Positive)* gives the proportion of coefficients that are significantly negative (positive) at a 5% level using autocorrelation and heteroscedasticity consistent standard errors (Newey and West, 1987). For means, the *t*-statistics for the test of zero mean are given in the parentheses. For the proportion of negatives and positives, the figures in the parentheses are *z*-statistics testing whether the proportion is equal to 2.5%, which it would be under no exposure to provision of liquidity. Figures significant at a 5% level are bolded. There are a total of 5,800 hedge funds in the sample.

	Mean	Negative	Positive
R_{LP}	0.064 (14.70)	3.3% (3.70)	7.7% (25.48)
Extreme, all stocks	-0.019 (-23.20)	14.9% (60.39)	2.7% (0.84)
R_{LP}	0.057 (12.84)	3.3% (3.87)	7.6% (24.64)
Extreme, non-financials	-0.003 (-4.43)	4.5% (9.84)	3.0% (2.61)
Extreme, financials	-0.028 (-31.72)	25.5% (112.03)	2.8% (1.26)

Table XVI. The effect of autoquote on hedge funds' liquidity supply

This table presents the summary statistics of the coefficients from a regression where hedge funds' monthly returns are regressed on the returns to long-short portfolios that make five-day investments identical to those in our time t returns from providing liquidity portfolio, but where the investments are made after skipping n days. In this way, we explore the timing of the hedge funds supply of liquidity. In these regressions, we use as controls the seven Fung and Hsieh (2004) risk factors and the Sadka (2006) liquidity shock. The results are presented separately for the time periods before and after the implementation of automated quote dissemination by the NYSE (1/1994-12/2002 and 6/2003-12/2011, respectively). *Mean* gives the average of the coefficients for all funds, *Negative* (*Positive*) gives the proportion of coefficients that are significantly negative (positive) at a 5% level using autocorrelation and heteroscedasticity consistent standard errors (Newey and West, 1987). For means, the t -statistics for the test of zero mean are given in the parentheses. For the proportion of negatives and positives, the figures in the parentheses are z -statistics testing whether the proportion is equal to 2.5%, which it would be under no exposure to provision of liquidity. Figures significant at a 5% level are bolded.

	1/1994 – 12/2002			6/2003 – 12/2011		
	Mean	Negative	Positive	Mean	Negative	Positive
0 days	0.100 (9.93)	1.5% (-2.87)	11.3% (24.36)	0.028 (6.31)	3.9% (6.16)	4.8% (9.77)
1 day	0.117 (10.51)	1.6% (-2.43)	11.4% (24.66)	0.055 (13.60)	2.8% (1.31)	6.6% (17.86)
2 days	0.081 (7.42)	2.4% (-0.34)	7.7% (14.24)	0.094 (22.52)	2.6% (0.26)	13.0% (45.08)
3 days	0.039 (3.85)	3.6% (3.08)	7.1% (12.76)	0.112 (24.64)	2.9% (1.69)	18.5% (68.96)
4 days	0.025 (2.74)	4.4% (5.16)	7.1% (12.76)	0.126 (25.40)	3.3% (3.59)	20.0% (75.53)
5 days	0.003 (0.31)	4.9% (6.65)	6.0% (9.63)	0.115 (22.55)	3.4% (3.78)	18.6% (69.63)
6 days	0.001 (0.12)	5.6% (8.44)	6.6% (11.27)	0.047 (9.43)	4.0% (6.44)	7.4% (21.10)
7 days	-0.011 (-1.29)	5.4% (7.99)	5.7% (8.74)	-0.019 (-4.51)	4.7% (9.30)	3.4% (3.78)
8 days	-0.021 (-2.28)	4.0% (4.27)	4.6% (5.91)	-0.042 (-11.43)	4.9% (10.25)	2.4% (-0.41)
9 days	-0.027 (-2.92)	3.8% (3.68)	4.5% (5.46)	-0.038 (-10.42)	5.8% (14.15)	2.9% (1.59)
10 days	-0.008 (-0.85)	3.0% (1.29)	5.0% (6.95)	-0.064 (-16.62)	8.7% (26.52)	2.2% (-1.26)

Figure 1. Monthly returns from our liquidity providing trading strategy

This figure presents the monthly returns from providing liquidity during our sample period 8/1990-12/2011. These are returns on a zero-investment long-short trading strategy in which five-day expected excess returns are used as portfolio weights when forming the long and the short portfolios for five-day investments, as in Equation (2) in the text. When constructing the portfolio, the expected returns are calculated using six month moving averages of coefficients for return reversal, as in Equation (1) in the text, until six days prior to taking positions. Portfolio returns are based on averages of the returns of all open positions. There is no consideration for transaction costs.

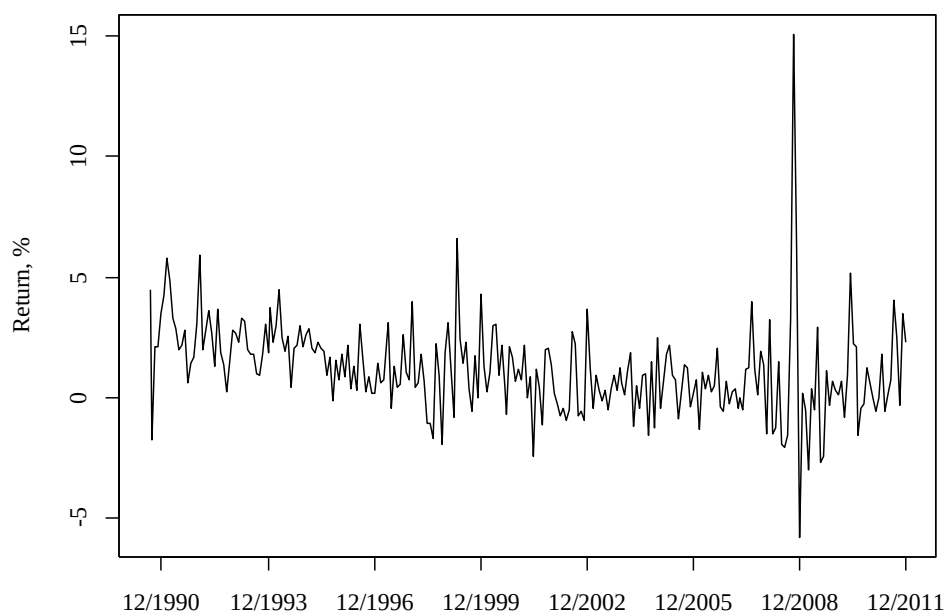


Figure 2. Time variation in hedge funds' provision of liquidity

This figure presents the average coefficient of the returns from providing liquidity, R_{LP} , conditional on conditioning variables, in a regression of hedge fund returns on the seven Fung and Hsieh (2004) risk factors, R_{LP} , liquidity shocks, and the interaction of R_{LP} with conditioning variables. The figure is based on results reported in column 3 of Table X. The conditioning variables are the one-month lagged liquidity level (Pastor and Stambaugh, 2003) and the change in TED spread (Treasury-over-Eurodollar), which proxies for the change in cost of leverage. The gray areas present the 95% confidence interval of the average coefficient.

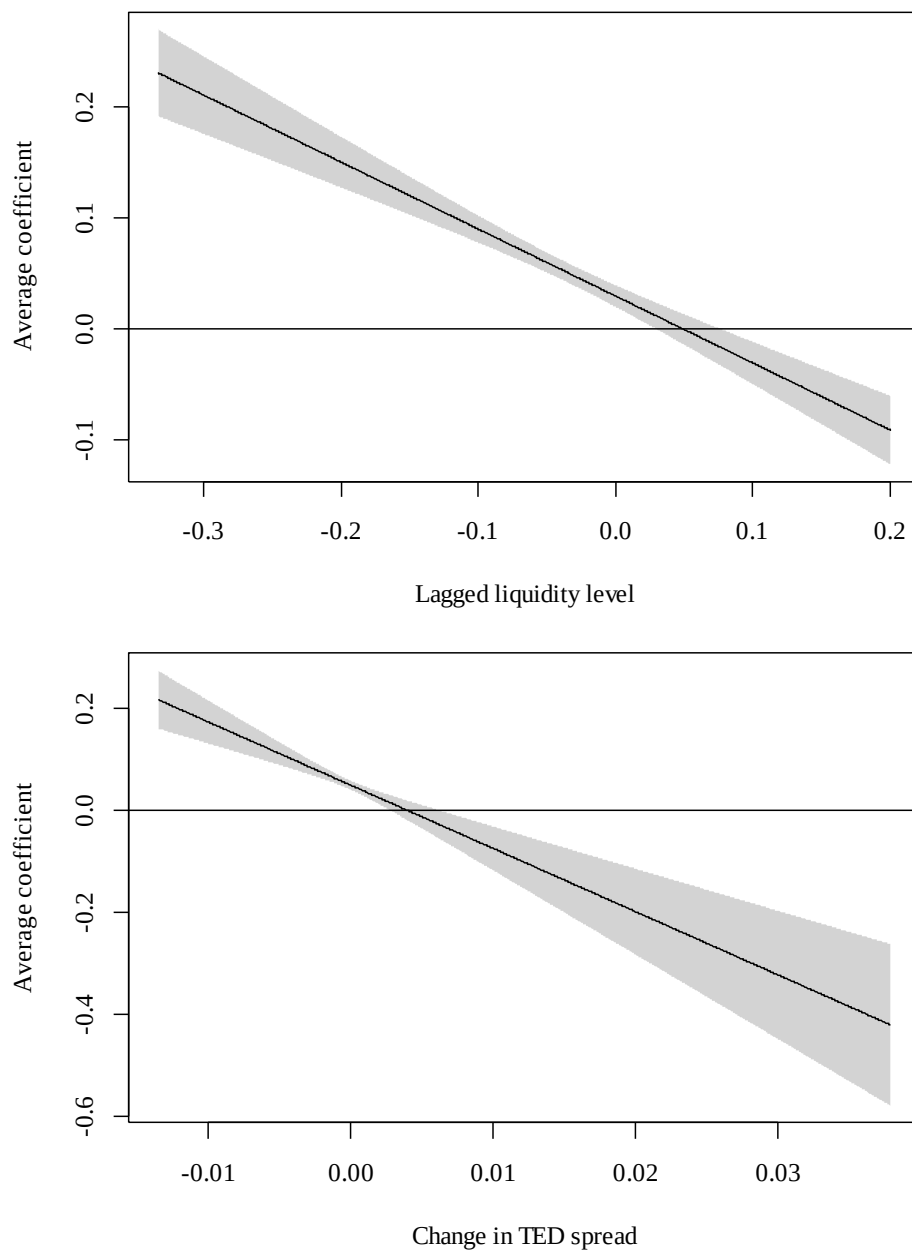


Figure 3. Time-varying proportions of liquidity suppliers and demanders

This figure shows the time-varying proportion of funds with statistically significant (at the 5% level) positive and negative loadings on the returns from providing liquidity, R_{LP} . The proportions are estimated using rolling 36-month windows. The horizontal lines mark the following events: 1. Asian crisis begins in 7/1997, 2. the collapse of Long-Term Capital Management in 9/1998, 3. beginning of automated trading in the NYSE in 1/2003, 4. quant crisis in 8/2007, 5. the bankruptcy of Lehman Brothers in 9/2008, and 6. the collapse of MF Global in 10/2011.

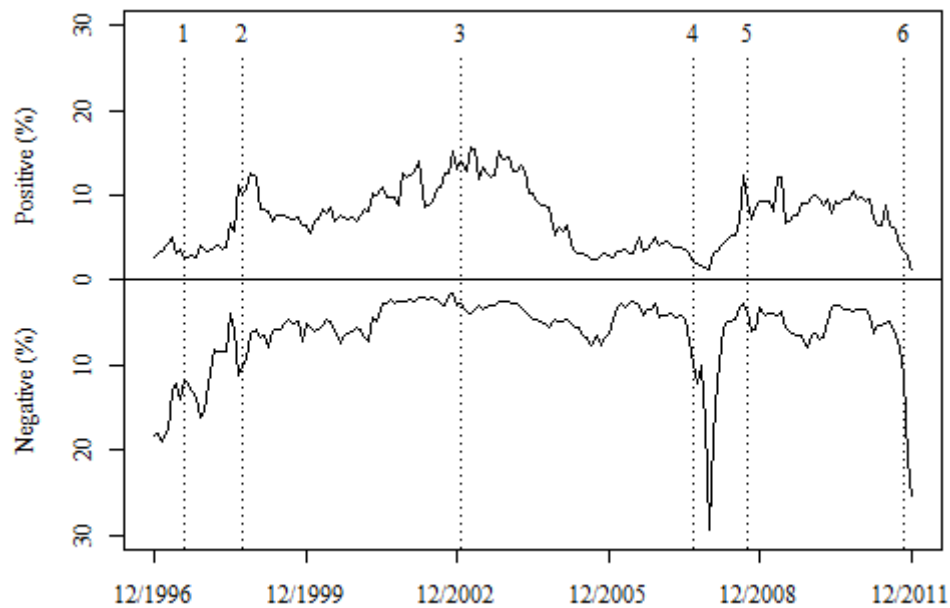


Figure 4. Monthly estimates of five-day return reversal

This figure shows the monthly estimates of five-day return reversal between 1/1994 and 12/2011. Our five-day return reversal estimate is the negative of the coefficient for the previous day's return, $-\hat{\beta}_t$, from daily regressions where stocks' five-day future excess returns are regressed on each of the stocks' past 20 day's excess returns. Excess returns are calculated relative to the corresponding equal-weighted Fama-French industry indexes.

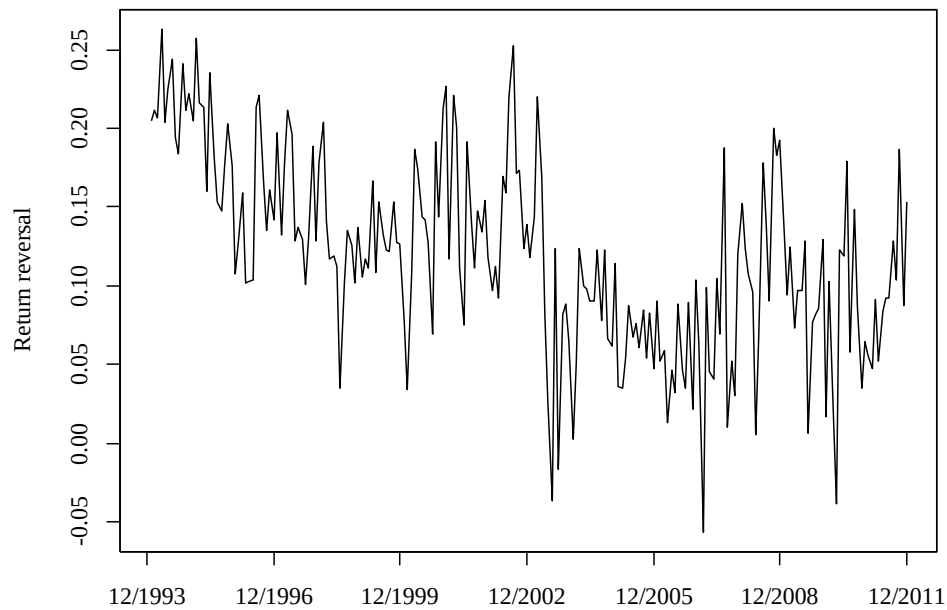


Figure 5. Time-varying proportions of liquidity suppliers and demanders in the extreme return stocks

This figure shows the time-varying proportion of funds with statistically significant (at the 5% level) positive and negative loadings on the returns from providing liquidity to the extreme return stocks. The proportions are estimated using rolling 36-month windows. The horizontal lines mark the following events: 1. Asian crisis begins in 7/1997, 2. the collapse of Long-Term Capital Management in 9/1998, 3. beginning of automated trading in the NYSE in 1/2003, 4. quant crisis in 8/2007, 5. the bankruptcy of Lehman Brothers in 9/2008, and 6. the collapse of MF Global in 10/2011.

